Formula CarMaker Lap Time Simulation Event 2025

Lap Time Simulation Competition Documentation

Version 1.0

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Introduction

The Lap Time Simulation (LTS) event within Formula student offers teams a chance to use IPG CarMaker to develop models of a Formula Student vehicle. These models will be submitted for a virtual competition, the details of which are in the FSUK 2025 rules.

Chapter 1

1 Getting Started

This document assumes that you already have CarMaker 13.1.1 installed and set up on your computer and have downloaded and set up the LTS CarMaker project from the IPG customer area. https://www.ipg-automotive.com/en/support/customer-area Please note, registration is required to access this area. Please see Section S7.1 in the FSUK 2025 rules for further information.

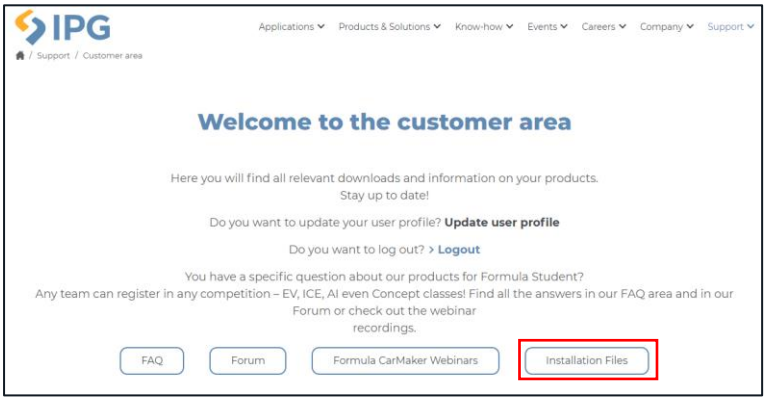

Figure 1.1 – IPG customer area

1.1 1.1 Selecting a Vehicle

In contrast to previous events, for LTS 2025, a starting point for each vehicle is provided in the LTS CarMaker Project. The vehicle that you choose does not have to be the same as your FS Class or Concept class entries. All vehicles have been through several rounds of Balance of Performance tuning, but some architectures may have more performance potential than others once fully developed. Testing multiple vehicles is highly recommended.

There are 2 main decisions to make when selecting a vehicle

1. Powertrain type
    - EV 1EM
    - EV 4EM
    - ICE Normally Aspirated
    - ICE Turbocharged

2. Aerodynamic concept
    - With Aero package
    - Without Aero package

Furthermore, for the rear wheel drive vehicles, there is the choice of differential

- Open differential
- Torsen differential
- Locked differential

All possible permutations of these vehicles are included in the installation package for a total of 20 vehicles. Your final submission must be based on one of these, with only changes explicitly listed in the rules.

1.2 1.2 Selecting Parts

Since most of the parts differences between the vehicles are already modelled in the 20 vehicles described in 1.1, the only parts that can be selected inside the GUI are the front and rear suspension designs, with different kinematics. The hardpoints of these designs are detailed in appendices 1 and 2.

There are five different designs for the front and another five for the rear. However, each of these designs has multiple options with adjustments for the static alignment.

IPG Kinematics files have been provided if you would like to look into these suspension hardpoints in more detail. The Model Check feature within CarMaker is also very useful when evaluating which parts to use.

1.3 1.3 Tunable parameters

CarMaker has tunable parameters known as Named Values (Nvalues). These parameters are numbers and represent parameterised physical characteristics of a part or system, for example the stiffness of a spring. As part of LTS, some of these Nvalues are open to tuning

- Suspension tunables (available to tune for front and rear suspensions)
    - Spring rate
    - ARB rate
    - Damper Low speed pull
    - Damper High speed pull
    - Damper Low speed push
    - Damper High speed push
    - Bumpstop push unstressed length
    - Bumpstop pull unstressed length
- Brake tunables – rear only
    - Rear brake pressure ratio
- Powertrain tunables
    - ICE only
        - Gear ratio 1
        - Gear ratio 2
        - Gear ratio 3
        - Gear ratio 4
        - Gear ratio 5
        - Gear ratio 6
    - EV 4EM only

            - Front motor ratio
        - EV 1EM only
            - Differential ratio
    - Driver parameters
        - Lateral acceleration time preview
        - Longitudinal acceleration time preview
        - Longitudinal braking time preview
        - Corner cutting coefficient
        - g-g diagram and exponents

Exact details of where these values are entered into the CarMaker GUI and the associated name in the vehicle infofile are included in Chapter 2.5.

Chapter 2

2 Developing your vehicle

2.1 2.1 Driver adaption

After changing vehicle, changing any parts or tunables as desired, the driver adaption must be run in order to teach IPG Driver how the vehicle behaves. Like any form of motorsport, the driver behaviour is a very important part of the overall vehicle performance.

A driver adaption will be required any time that you make a change to the vehicle, no matter how small. In the LTS final competition, the driver adaption will be run for each submission, giving the driver up to date knowledge. Testing with a driver with out-of-date knowledge will almost certainly lead to different results.

To start the driver adaption, open the Simulation menu and then select Driver Adaption as shown in Figure 2.1

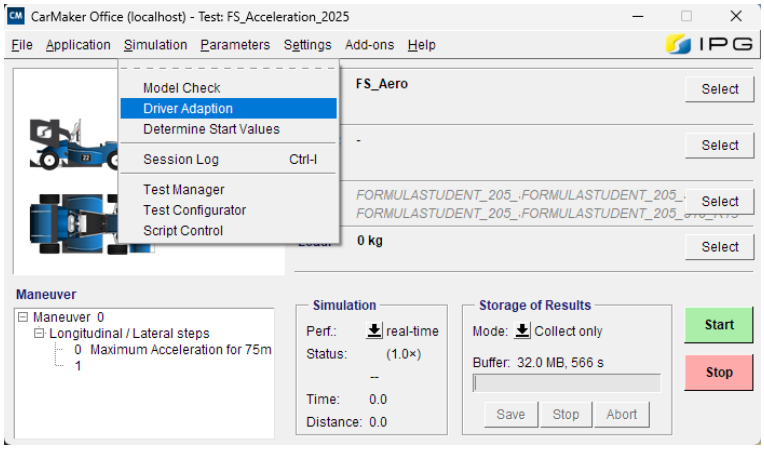

Figure 2.1 – Main GUI Simulation Menu

This will open the Driver Knowledge Editor shown in Figure 2.2

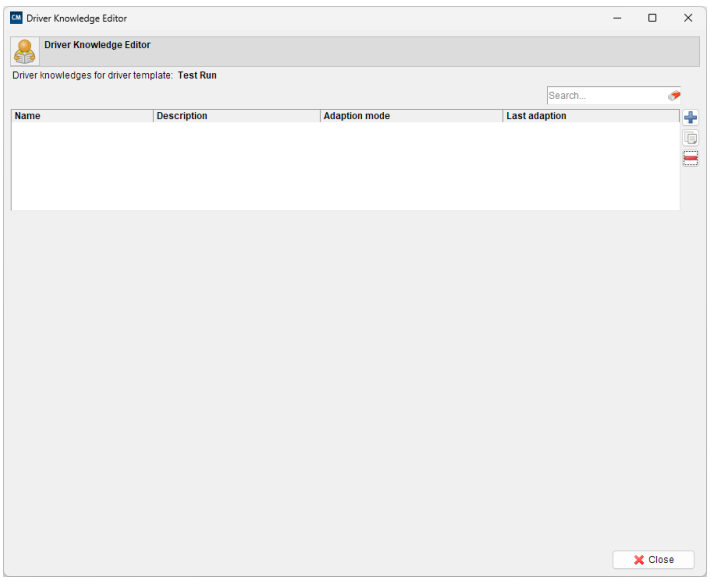

Figure 2.2 – Driver Knowledge Editor

The first time you do this, you will need to add driver knowledge using the blue plus icon on the right of the screen.

Once you have added driver knowledge, you must make sure that the Vehicle and Basic Driver Adaption parameters are as per figure 2.3. Note, for the purposes of LTS, even if using the EV 4EM powertrain, the driver must be configured as RWD.

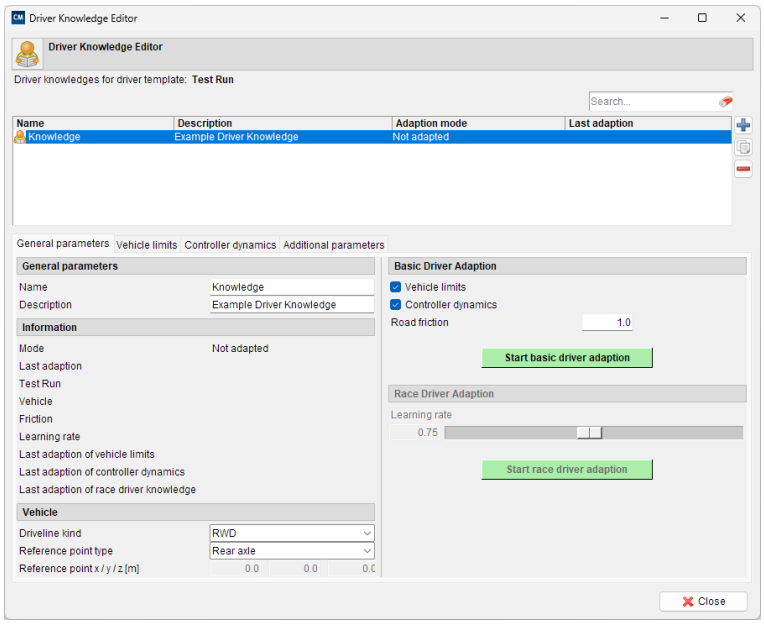

Figure 2.3 – Driver Knowledge parameters

You may now start the basic driver adaption. It is recommended to change the Simulation Speed to be faster than real-time in order to finish the Driver Adaption more quickly, since it is a long test run. You can do this from the main GUI, using the Perf: drop down option as shown in figure 2.4

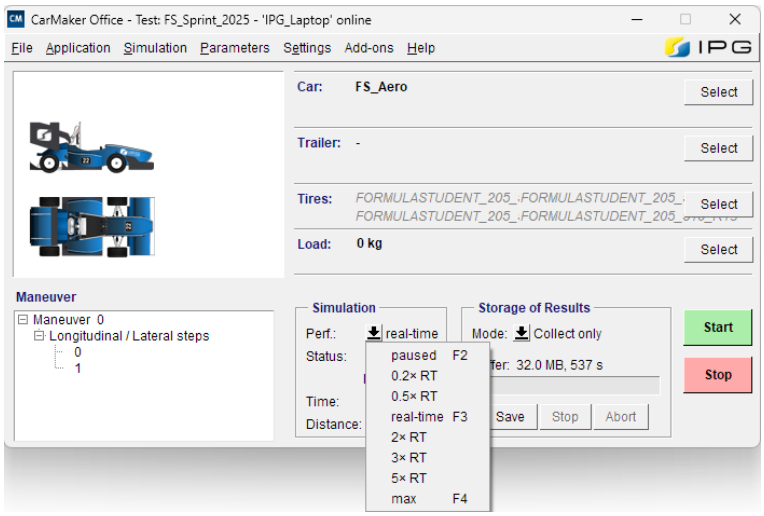

Figure 2.4 – Simulation Speed menu

After the driver adaption is complete, the driver knowledge must be selected in the Ego vehicle parameters. **This must also be done after making any changes to the driver knowledge**, such as tuning the Driver Lateral preview Time.

Access these by selecting Parameters > Manuever to open the Manuever settings, then change to Vehicle Parameters tab and select the appropriate knowledge in the IPG Driver section by using the button highlighted in Figure 2.5

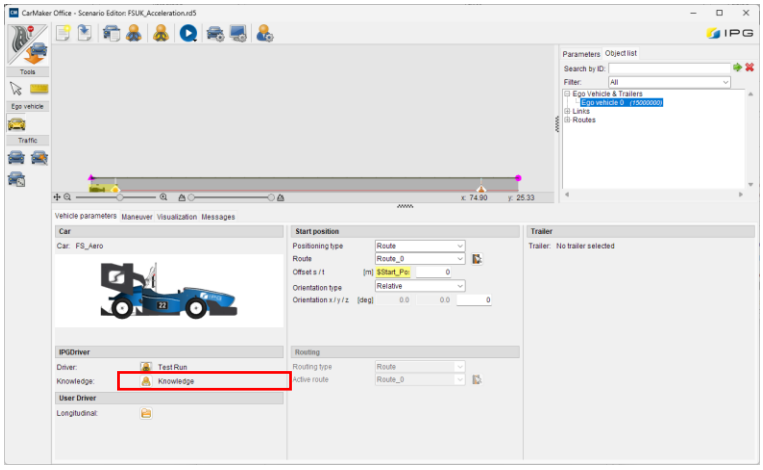

Figure 2.5 – Driver Knowledge selection

Now that the driver has knowledge, the driver must be told to act as a racing driver. Set this in the Driver Template Editor by selecting Parameters > Driver and set the Mode to Race driver, as shown in figure 2.6.

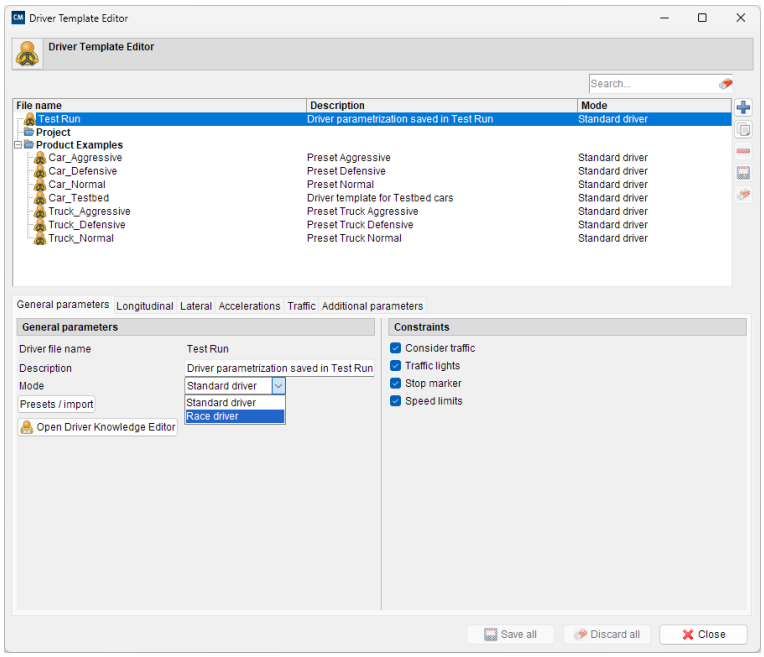

Figure 2.6 – Race drive Mode selection.

Please note, this process will have to be performed each time you make any change to the vehicle or change the test run.

2.2 2.2 Driver Tuning

IPG Driver has a vast number of parameterisations to be able to replicate real driver behaviours. As part of the LTS event, there are a limited number of Driver parameters you may tune. These can be broken down into Driver parameters and Driver Knowledge Parameters.

The tunable Driver parameters are

- Corner cutting coefficient (Carmaker infofile name *Driver.Course.CornerCutCoef*)
- g-g Diagram (*Driver.Acc.GGExp*)

These can be found in the Driver Template Editor, under the Lateral and Accelerations tabs. Please note, for the g-g-Diagram, you must set the speeds to 40 and 80km/h as per Figure 2.7, the values in the 4 highlighted cells are free to tune.

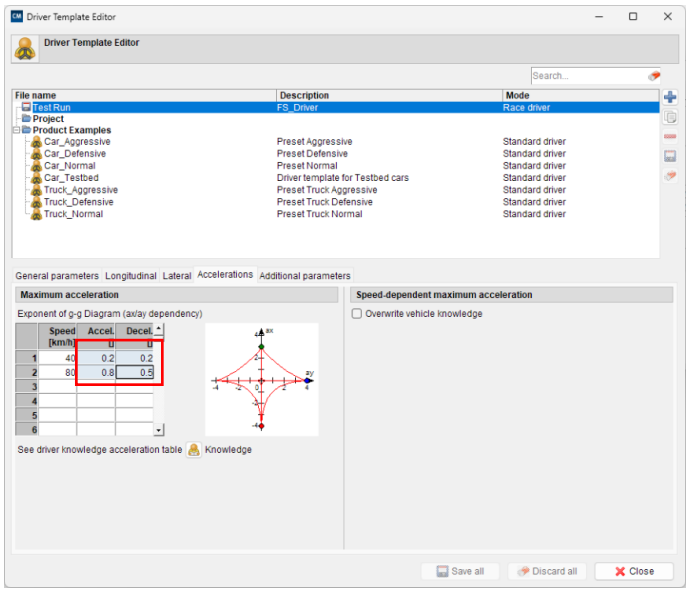

Figure 2.7 – Driver g-g Diagram

The tunable Driver Knowledge Parameters are

- Longitudinal Preview time acceleration (Carmaker infofile name *Knowl.Long.tPreviewAcc*) Determines the reference point for throttle control.
- Longitudinal Preview time braking **(***Knowl.Long.tPreviewBra***)** Determines the reference point for braking control.
- Lateral preview time (*Knowl.Lat.tPreviewAcc)* Determines the reference point for steering control.

These can be found in the Driver Knowledge Editor, under the Controller dynamics tab as shown in figure 2.8

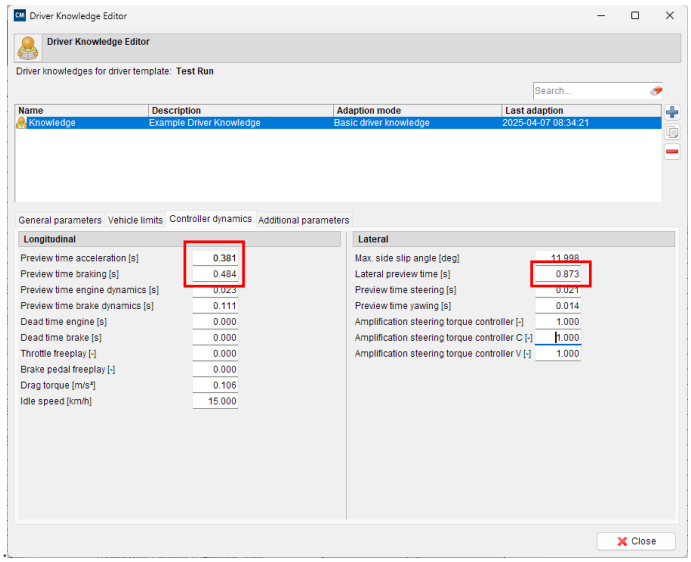

Figure 2.8 – Driver Knowledge tunable Parameters

After changing any of the Driver Knowledge values, you will need to select the appropriate Driver Knowledge as per Figure 2.5.

If the driver is not able to finish the Sprint Lap and goes out of the track, changing Knowl.Lat.tPreview might help. The combination of this parameter and the Cutting Corner Coefficient has a significant effect on the number of cones hit

2.3 2.3 Vehicle Selection

To select a vehicle, click the Select button in the Car section as shown in figure 2.9.

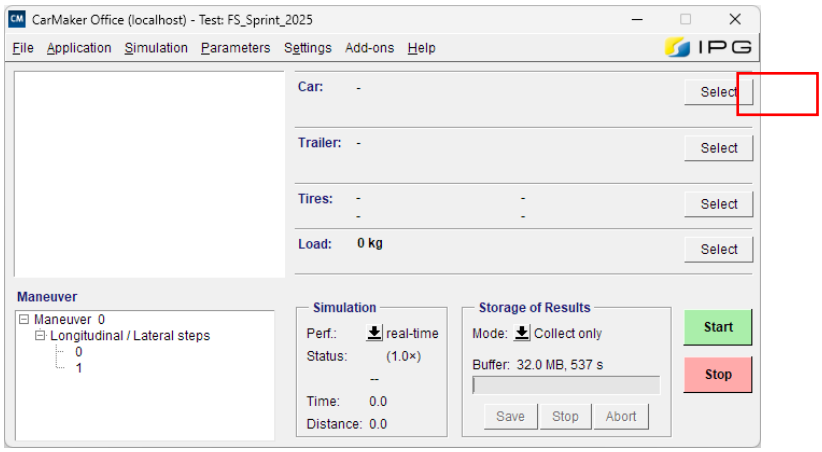

Figure 2.9 – Car selection in the main GUI

The starting vehicles for LTS are saved in the folder Baseline_Vehicles in the project directory as shown in figure 2.10. If would like to make any changes, it is recommended to create a copy of this in the project directory, so that you can always return to a baseline vehicle if needed.

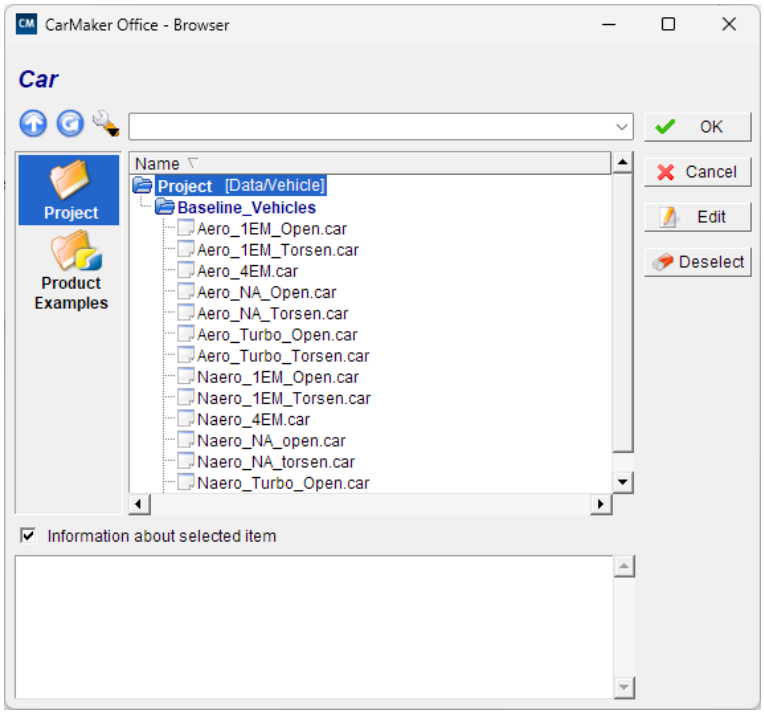

Figure 2.10 – Baseline vehicle folder

2.4 2.4 Vehicle Parts changes

As explained in Chapter 1, the only parts requiring user changes during LTS competition are the front and rear suspension kinematics. These can be changed by navigating to Parameters > Car and selecting the suspension tab. The kinematics option is visible here, as per Figure 2.11. Clicking the Model drop down menu gives you the option to change the suspension parameterisation file (.skc).

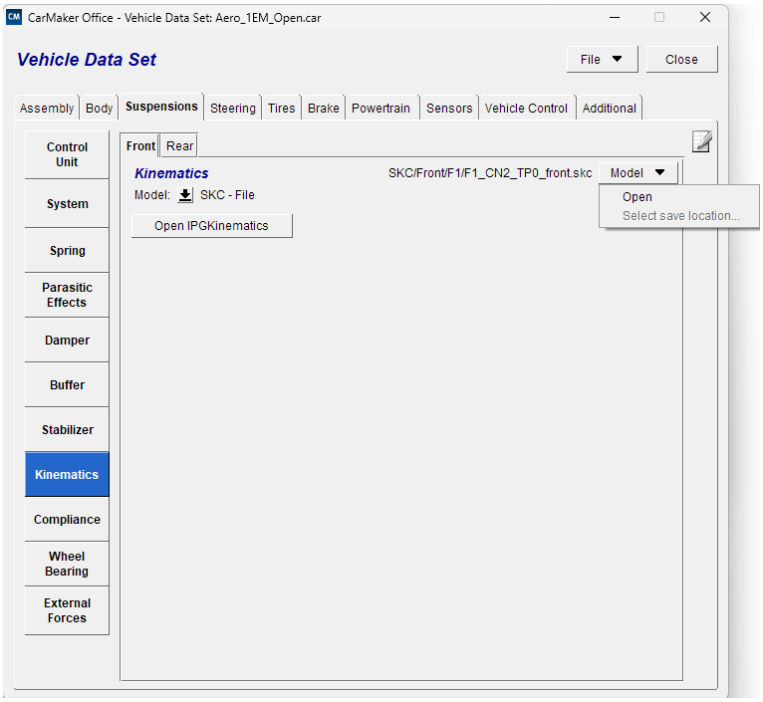

Figure 2.11 – Suspension kinematics model change

2.5 2.5 Vehicle tuning parameter (Nvalue) changes

Nvalue changes are simple to make either through the GUI or through the vehicle infofile. The variables that are allowed to be tuned are as below. Please note the exact name and order of these variables, since these will have to be filled into the submission form.

Spring rate, Carmaker infofile name *SuspF.Spring* and *SuspR.Spring*

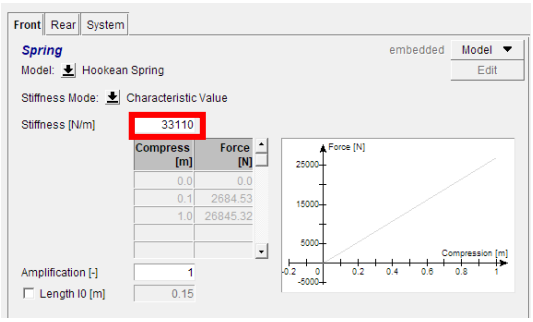

Figure 2.12 – Spring rate GUI entry

ARB rate, *SuspF.Stabi* and *SuspR.Stabi*

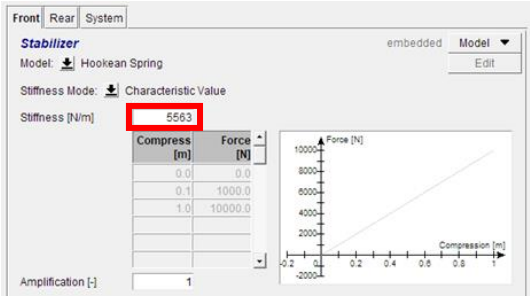

Figure 2.13 – ARB rate GUI entry

Damper rates, please note, the velocity values are fixed

- Damper Low speed push, *SuspF.Damp_Push, SuspR.Damp_Push*
- Damper High speed push, *SuspF.Damp_Push, SuspR.Damp_Push*
- Damper Low speed pull, *SuspF.Damp_Pull. SuspR.Damp_Pull*
- Damper High speed pull, *SuspF.Damp_Pull. SuspR.Damp_Pull:*

Figure 2.14 – Damper rate GUI entry

Bumpstop push unstressed length, *SuspF.Buf_Push.tz0, SuspR.Buf_Push.tz0*
Bumpstop pull unstressed length *SuspF.Buf_Pull.tz0, SuspR.Buf_Pull.tz0*

Figure 2.15 – Bumpstop unstressed length GUI entry

Rear brake torque to pressure ratio (only one value, both left and right must be the same), *Brake.System.pWB2Trq*

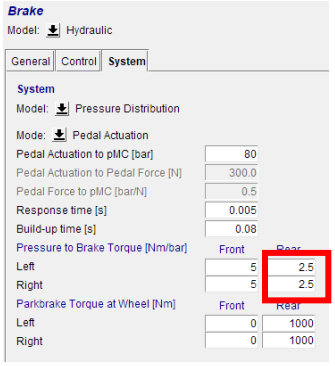

Figure 2.16 – Rear brake torque pressure ratio GUI entry

ICE only, Gear ratios 1-6, *PowerTrain.Gearbox.0.iForwardGears*

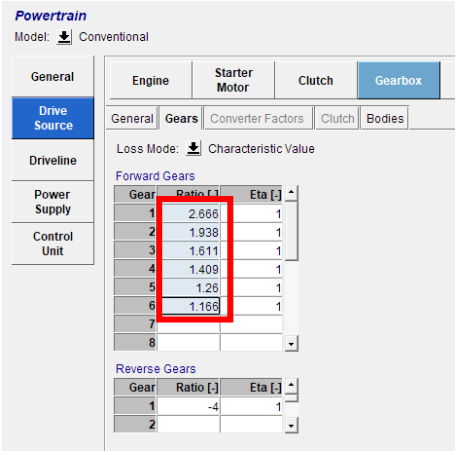

Figure 2.17 – ICE only gear ratio GUI selection

EV 4EM only, Electric motor front ratio, *PowerTrain.Motor.Ratio* and *PowerTrain.Motor1.Ratio* (only one value, Drive Source 0 and Drive Source 1 must be the same)

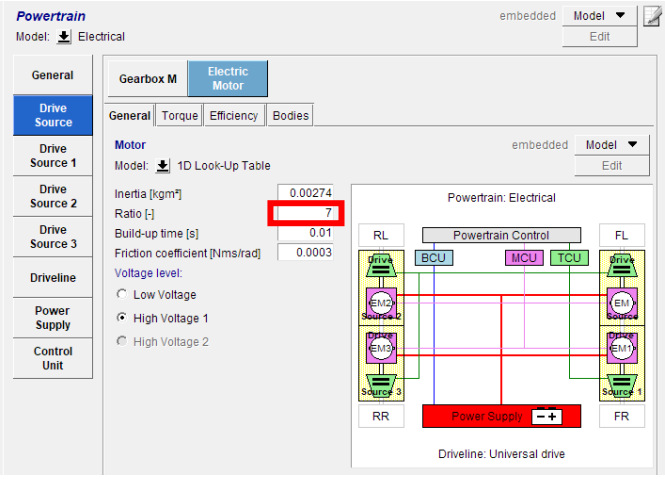

Figure 2.18 – EV 4EM front motor ratio

EV 1EM only, Differential ratio

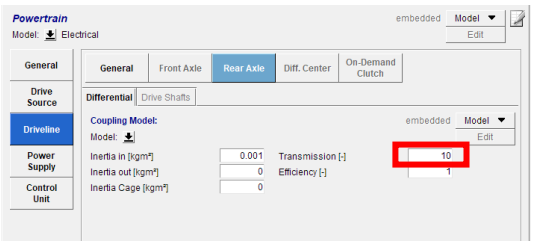

Figure 2.19 – EV 1EM rear differential ratio

Chapter 3

3 Entering your vehicle for the competition

Once you have finished developing your vehicle and wish to submit your final submission, please fill your vehicle data in the Excel sheet provided, LTS_Submission_Form_v*X*.xlsx. Please see Chapter 2 if you are unclear about exactly which values to enter. All units are to be the default units as displayed in the GUI.

You are only permitted to enter data in the blue cells. Any data entered outside of these will not be considered in the virtual competition. Please note that for guidance, example data is already filled in, this must be overwritten with your entry details, otherwise this example data will be used!

The Data check category is not used for scoring, but may be used in the event of investigating a failed run or other issue. In the competition, every test run will be calculated from scratch using the parameters provided in the Excel sheet.

Once you have filled in your entry form, it is advisable to start from the original vehicle, changing values via the GUI to recreate your final submission, to verify that this creates the intended vehicle. The competition will be run with what is submitted on the sheet and any inputs that lead to non-completion of events will be scored accordingly.

Please save this form as your teamnumber.xlsx, (for example, Team 8 would save this as 8.xlsx) Then log on to your FS Team Portal and upload this file.

4 Appendix 1: Front Suspension Config Hardpoints

	Vehicle - Bushing Front Upper Wishbone (A)			Vehicle - Bushing Rear Upper Wishbone (B)			Subframe - Bushing Front Lower Wishbone (C)			Subframe - Bushing Rear Lower Wishbone (D)		
	x	*y*	*z*	*x*	*y*	*z*	*x*	*y*	*z*	*x*	*y*	*z*
F1	-44.0	-234.0	320.0	227.0	-234.0	320.0	-46.0	-144.0	128.0	292.0	-144.0	128.0
F2	-44.0	-234.0	310.0	227.0	-234.0	310.0	-46.0	-144.0	138.0	292.0	-144.0	138.0
F3	-44.0	-234.0	330.0	227.0	-234.0	330.0	-46.0	-144.0	118.0	292.0	-144.0	118.0
F4	-44.0	-234.0	330.0	227.0	-234.0	330.0	-46.0	-144.0	138.0	292.0	-144.0	138.0
F5	-44.0	-234.0	310.0	227.0	-234.0	310.0	-46.0	-144.0	118.0	292.0	-144.0	118.0

5 Appendix 2: Rear Suspension Config Hardpoints

	Vehicle - Bushing Front Upper Wishbone (A)			**Vehicle - Bushing Rear Upper Wishbone (B)**			**Subframe - Bushing Front Lower Wishbone (C)**			**Subframe - Bushing Rear Lower Wishbone (D)**		
	x	*y*	*z*	*x*	*y*	*z*	*x*	*y*	*z*	*x*	*y*	*z*
1	-646.0	-300.0	299.0	-110.0	-235.0	299.0	-670.0	-144.0	125.0	-110.0	-180.0	125.0
R2	-646.0	-300.0	289.0	-110.0	-235.0	289.0	-670.0	-144.0	135.0	-110.0	-180.0	135.0
R3	-646.0	-300.0	309.0	-110.0	-235.0	309.0	-670.0	-144.0	115.0	-110.0	-180.0	115.0
R4	-646.0	-300.0	309.0	-110.0	-235.0	309.0	-670.0	-144.0	135.0	-110.0	-180.0	135.0
R5	-646.0	-300.0	289.0	-110.0	-235.0	289.0	-670.0	-144.0	115.0	-110.0	-180.0	115.0